3. Part A — Research and Analysis (Multi-Level Perspective)

3.1 Namibian Wireless and Wide Area Network (WAN) Landscape Overview of Market Operators and Regulatory Structure

Namibia's wireless and Wide Area Network (WAN) sector consists of several companies that provide mobile, internet, fiber, and other communication services. The sector is regulated by the **Communications Regulatory Authority of Namibia (CRAN)**, which was created under the Communications Act of 2009.

CRAN is responsible for managing the use of communication frequencies (spectrum), making sure companies share infrastructure fairly, promoting competition, and helping improve communication services in rural areas.

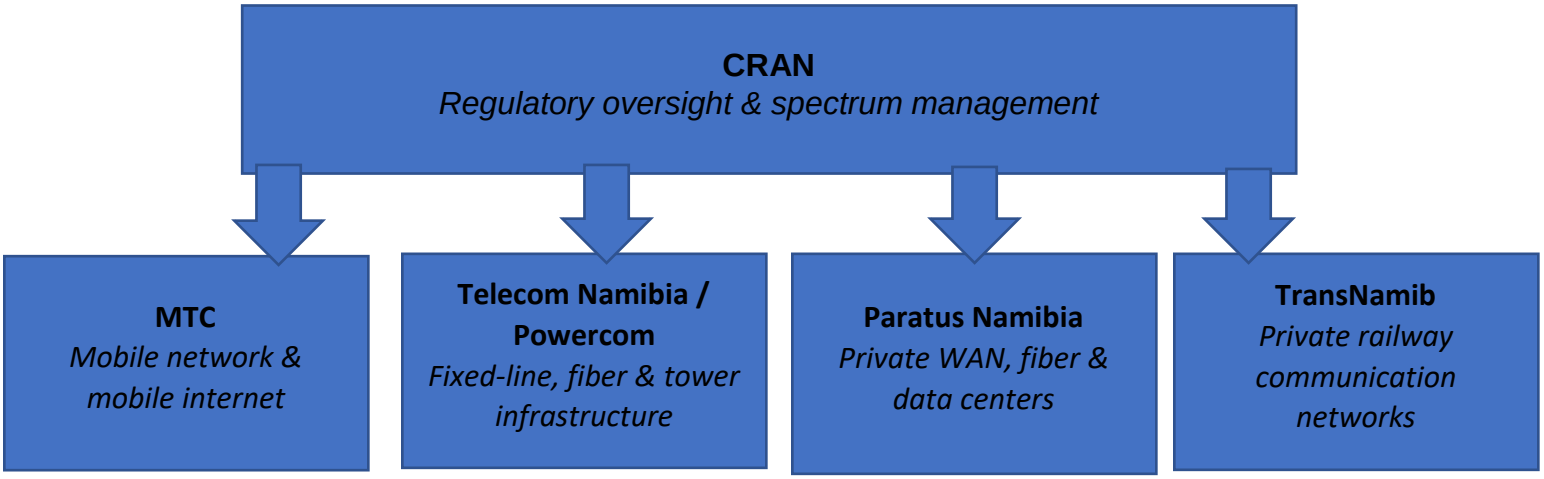
For many years, companies linked to the government have been major providers in Namibia's telecommunications sector. **Mobile Telecommunications Limited (MTC)** is the largest mobile network provider and has a large share of the country's telecommunications infrastructure. MTC mainly focuses on mobile communication services, cellular networks, and mobile internet.

Telecom Namibia, together with its tower infrastructure subsidiary **Powercom**, is one of the main providers of fixed-line communication services. It operates a large fiber-optic network and provides services such as fixed internet and Wi-Fi connectivity.

A major private competitor is **Paratus Namibia**, which focuses strongly on business and enterprise WAN services. The company operates private fiber networks and data centers and provides high-speed international connectivity through the Equiano subsea cable. This helps businesses in Namibia connect more efficiently to international networks.

Some organizations also operate their own private communication networks for specific purposes. For example, **TransNamib** uses private fiber and wireless networks along railway routes. These networks support railway communication, signaling, cargo tracking, and other railway operations without depending entirely on public telecommunications networks.

STRUCTURE



4G/5G Rollout Status and the Digital Divide

Namibia is working to improve its broadband and mobile network infrastructure by moving from older technologies such as 2G and 3G to faster 4G and 5G networks. This development is supported by CRAN's 2025–2027 Spectrum Assignment Strategy. From 2026, older 2G and 3G networks are planned to be gradually reduced so that their frequencies can be used for faster and more efficient 4G and 5G services.

Mobile network operators are investing large amounts of money to support this development. In the first quarter of 2026, telecommunications investment reached about N$411 million. Telecom Namibia has also planned to build 500 new mobile base stations by 2027, with the goal of improving 4G and 5G coverage, especially in remote areas. MTC is also investing in upgrading its network and expanding its communication towers.

However, Namibia still has a significant **digital divide between urban and rural areas**. In major towns such as Windhoek, Swakopmund, and Oshakati, people generally have better access to high-speed internet, fiber connections, 4G, and 5G services. In contrast, many people living in rural and remote areas have limited access to mobile networks and internet services. National census information indicates that almost 60% of people in rural areas do not own a mobile phone, while cellular device ownership across the country is approximately 28.5%.

To reduce this gap, the Namibian government has allocated **N$140 million over three years** to the Universal Service Fund managed by CRAN. The fund is intended to improve telecommunications services in areas that have limited network coverage. An initial **N$31 million** was provided to MTC to construct new transmission towers in regions where 4G coverage is below 80%. These areas include **Kavango West, Kavango East, Kunene, Ohangwena, and Oshikoto**.

Overall, Namibia is making progress in expanding 4G and 5G networks, but more investment is still needed to ensure that people in rural and remote communities receive the same level of digital access as those living in major urban areas.

Geographic Constraints: Terrestrial vs. Satellite WAN Cost Models

Namibia's large size and low population density have a major impact on how Wide Area Networks (WANs) are designed and deployed. The country covers about **825,615 square kilometers** but has a population of fewer than three million people. Because communities are often far apart, connecting them using traditional fiber-optic cables or microwave links can be difficult and expensive. In many cases, network infrastructure has to cross long distances of desert and other remote areas to reach a small number of users.

Traditional terrestrial networks can therefore be very costly to build and maintain. A mobile tower or wireless relay station may provide coverage over a limited area, meaning that many towers or relay stations may be needed to connect remote locations. Building and maintaining these sites over large, unpopulated areas requires significant investment. As a result, terrestrial WAN connections are generally more practical in cities and areas where many users are located close together.

Feature	Terrestrial WAN (Fiber/Microwave)	Satellite WAN (VSAT/LEO)
Capital Cost	Very high because of the cost of laying cables and building network infrastructure	Lower because less physical infrastructure is required
Latency	Generally low, especially with fiber connections	VSAT has high latency, while LEO provides much lower latency
Geographic Coverage	Limited by the availability of physical network infrastructure	Can provide coverage in remote and isolated areas
Best Use	Cities, towns, and areas with high population density	Remote communities, mines, farms, and industrial sites

Because of these geographical challenges, **satellite communication has become an important option for providing connectivity in remote parts of Namibia**. Traditional **Very Small Aperture Terminal (VSAT)** systems have been used for many years to provide internet and communication services in areas where terrestrial networks are difficult to install. However, VSAT systems normally have higher latency because they use satellites located in geostationary orbit, which are very far from Earth.

More recently, **Low Earth Orbit (LEO)** satellite services, such as Starlink, have become an alternative. LEO satellites operate much closer to Earth, generally at several

hundred to around 1,200 kilometers above the surface. Their shorter distance allows them to provide significantly lower latency than traditional VSAT systems. They also reduce the need to build expensive fiber cables and relay towers across large, remote areas.

Therefore, Namibia's WAN infrastructure is likely to continue using a **combination of terrestrial and satellite technologies**. Fiber and microwave networks are more suitable for cities and populated areas, while satellite solutions can provide reliable connectivity to remote communities and isolated industrial locations.

Enterprise WAN Topologies Across Key Namibian Sectors

Different organizations in Namibia use different WAN technologies depending on their location, size, and communication needs. The main examples include banking, mining, logistics, and education.

1. Banking and Financial Institutions

Banks such as Bank Windhoek and First National Bank Namibia require secure and reliable networks to connect their branches, ATMs, and headquarters. Their main offices in Windhoek often use high-speed private fiber connections and data centers. Branches in towns such as Walvis Bay and Swakopmund may use MPLS connections provided by companies such as Telecom Namibia or Paratus.

For backup purposes, especially at rural branches and ATMs, banks can use encrypted satellite connections such as VSAT. This provides an alternative connection if the main fiber or terrestrial network fails, allowing important banking services and transactions to continue.

2. Mining and Extraction Companies

Mining companies often operate in remote areas where normal telecommunications infrastructure is limited or unavailable. Because of this, they may use satellite communication and private LTE or 5G networks to connect mining sites with their main offices.

These networks allow mining companies to monitor vehicles and equipment, track employees for safety purposes, transfer large amounts of data, and manage mining operations remotely. Reliable connectivity is especially important because mining sites can be located far from major towns and cities.

3. Logistics and Supply Chain Companies

Logistics companies use technologies such as **Software-Defined Wide Area Networking (SD-WAN)** and **IPsec VPNs** to securely connect their different offices, warehouses, and transport operations.

For example, a logistics company can connect warehouses at the Port of Walvis Bay with its offices in Windhoek using secure internet connections. This can be cheaper than using dedicated leased lines. Companies can also use mobile networks to track trucks and other vehicles while they are travelling between locations.

4. Academic Institutions and Universities

Universities such as the **University of Namibia (UNAM)** and the **Namibia University of Science and Technology (NUST)** use WAN infrastructure to connect their different campuses and facilities.

High-speed fiber connections allow universities to share resources between campuses and support services such as online learning, video conferencing, centralized databases, and access to academic systems. These networks also make it easier for students and staff at different campuses to access the same digital resources.

Summary

Overall, WAN requirements differ from one sector to another. **Banks** focus mainly on security and reliability, **mining companies** require connectivity in remote locations, **logistics companies** need affordable and flexible connections, while **universities** require high-speed networks for sharing academic resources. Therefore, organizations often combine technologies such as **fiber, MPLS, SD-WAN, VPNs, cellular networks, and satellite communication** to meet their specific networking requirements.

3.2 SADC Regional Perspective: Infrastructure and Policy Maturity Matrix

When comparing WAN and wireless network development across the **Southern African Development Community (SADC)**, there are noticeable differences between Namibia and countries such as South Africa and Botswana. These differences are mainly related to infrastructure, geography, market size, and access to international internet connections.

South Africa is one of the main telecommunications hubs in Africa. It has several major undersea cables that connect the country to international networks, including **WACS, SAFE, Seacom, and 2Africa**. Having many submarine cables gives South Africa access to large amounts of international bandwidth and helps reduce the cost of connecting to global networks.

Namibia is also connected to international networks through the **WACS submarine cable**, which reaches the country through the Swakopmund landing station. In the past, Namibia experienced relatively high internet costs because its telecommunications

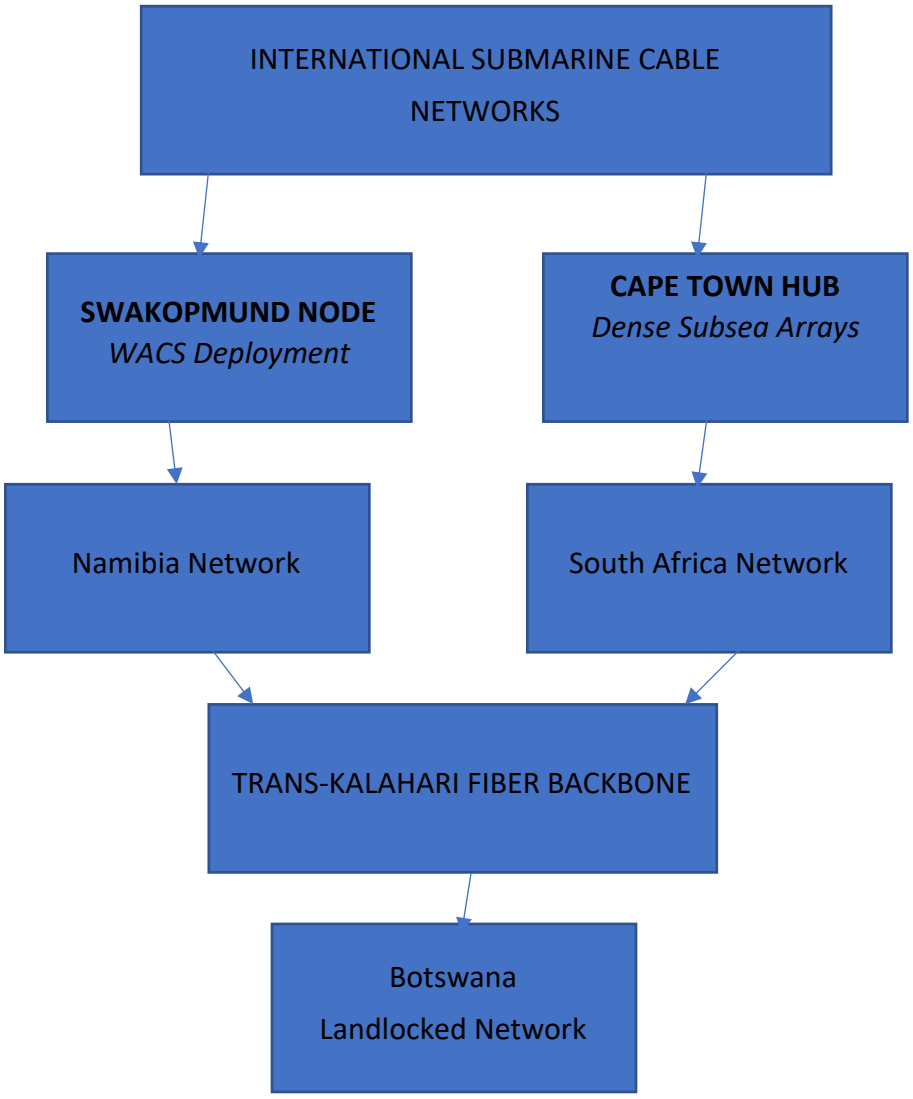
market is smaller than South Africa's. However, increased competition and investment by private companies have helped improve Namibia's fiber infrastructure and create stronger national network connections.

Botswana faces some similar challenges to Namibia because of its large land area and relatively low population density. However, Botswana is a **landlocked country**, meaning that it does not have direct access to submarine cable landing stations. As a result, it depends on neighboring countries such as South Africa and Zimbabwe to connect to international submarine cable networks.

Namibia has an advantage because it has access to the **Atlantic Ocean**. This allows the country to act as a potential telecommunications gateway for landlocked SADC countries. Through cross-border fiber connections, Namibia can help transport international data traffic between submarine cables and neighboring countries.

Regional Connectivity Structure

The basic regional connection can be represented as follows:



Summary

Overall, **South Africa has the most developed international telecommunications infrastructure** in the region because of its many submarine cable connections and large telecommunications market. **Namibia has a strategic advantage because of its coastline and access to WACS**, allowing it to connect to international networks and potentially provide connectivity to landlocked SADC countries. **Botswana, on the other hand, depends more heavily on cross-border connections through neighboring countries** because it has no direct access to submarine cable landing stations.

These differences show that geography, infrastructure investment, and regional cooperation play an important role in determining the cost, reliability, and availability of WAN and wireless services across the SADC region.

3.3 Global Perspective

Global Enterprise Case Study: DHL Global Supply Chain Network

To compare Namibia's WAN development with international practices, this project looks at the network infrastructure used by large global logistics companies such as **DHL Global Supply Chain**. Large logistics companies operate across many countries and require reliable networks to connect warehouses, offices, transport vehicles, tracking systems, and customers.

These companies commonly use **Software-Defined Wide Area Networking (SD-WAN)** together with a combination of private networks such as MPLS and public internet connections. They may also use cloud and edge-computing technologies to process information closer to where it is generated. For example, tracking devices, automated warehouses, and transport offices can send information to nearby network locations before the data is transferred to central cloud systems.

This type of hybrid network helps prevent a problem at one location from affecting the entire organization's operations. A similar approach could be used by **Kalahari Logistics Ltd**. The company could combine reliable local connections, such as fiber and VPNs, with high-speed point-to-point links. This would allow the company to expand its network while reducing the cost of depending entirely on expensive traditional leased connections.

Alignment with Global Standards

To ensure that the network can work effectively with modern technologies and international systems, the proposed design follows standards developed by major international technology organizations.

- **IEEE 802.11 – Wireless LAN:** Local offices and facilities can use modern Wi-Fi standards such as **Wi-Fi 6 (802.11ax)** and **Wi-Fi 7 (802.11be)**. These technologies are designed to provide reliable wireless connections when many devices are connected at the same time. Features such as MU-MIMO and OFDMA help improve network performance in busy warehouses and offices.
- **3GPP – Cellular and 5G Networks:** Cellular connections can follow 3GPP standards for 4G and 5G networks. These standards support modern features such as network slicing, which can help separate business traffic from normal public network traffic and improve security and reliability.
- **ITU-T – Telecommunications Standards:** The network can also follow ITU-T recommendations for fiber-optic communication and IP-based networks. These standards help ensure reliable data transmission and support important network requirements such as Quality of Service (QoS).

Maturity Benchmark: Namibia/SADC Compared with Global Trends

Globally, enterprise networks are increasingly moving toward technologies such as **Standalone 5G (5G SA), network slicing, Zero Trust security, SD-WAN, cloud networking, and Low Earth Orbit (LEO) satellite systems**. These technologies provide organizations with greater flexibility, security, and access to reliable connectivity.

Compared with Europe and North America, Namibia and the wider SADC region are still developing their enterprise networking infrastructure. Many organizations continue to depend on traditional fiber, microwave, and other physical backhaul systems. In addition, many 5G deployments in the region are still based on **Non-Standalone 5G (5G NSA)**, which uses existing 4G infrastructure.

However, the introduction of **LEO satellite services such as Starlink** is creating new opportunities for businesses in Namibia and other SADC countries. These services can provide high-speed internet to locations where installing fiber or other terrestrial infrastructure would be expensive or difficult. This allows companies operating in remote areas to adopt modern connectivity solutions without waiting for extensive terrestrial network expansion.

Overall, Namibia and the SADC region are gradually moving toward global networking standards. Although there is still a gap between local and highly developed international markets, new technologies such as **5G, SD-WAN, cloud networking, and LEO satellites** are helping to reduce this gap.